

Scenario 1: ATM Card Verification

A bank stores valid ATM card numbers.

Given a card number, check whether it exists.

Input

5

1001

1002

1003

1004

1005

1003

Output

Card Valid

Hint

Array Sorted → Binary Search

Scenario 2: Employee Login System

Company employee IDs:

101

205

150

300

250

Employee enters ID:

300

Check whether employee exists.

Output

Login Successful

Hint

Array Unsorted → Linear Search

Scenario 3: Online Exam Registration

Registered student IDs:

100
200
300
400
500

Student enters:

350

Output

Student Not Registered

Hint

Binary Search

Scenario 4: Flight Passenger Check

Passengers booked tickets:

501
205
703
101
402

Search passenger ticket:

703

Output

Passenger Found

Hint

Linear Search

Scenario 5: E-Commerce Product Search

Products:

1001

1002

1003

1004

1005

Search:

1004

Output

Product Available

Hint

Binary Search

Scenario 6: Hospital Patient Search

Patient IDs:

120

140

160

180

200

Search:

180

Output

Patient Record Found

Interview Twist

Ask:

Why Binary Search?

Answer:

Patient IDs are stored in sorted order.

Scenario 7: Cricket Team Selection

Player jersey numbers:

7

18

45

10

33

Find player:

10

Output

Player Selected

Hint

Linear Search

Scenario 8: Company Asset Tracking

Asset IDs:

100

200

300

400

500

600

700

800

Search Asset ID:

700

Output

Asset Found at Position 6

Hint

Binary Search

